

GRAND MARAIS AP, MN, USA

WMO: 727454

Lat: 47.838N

Lon: 90.383W

Elev: 548

StdP: 94.91

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.0	-26.0	-32.9	0.2	-26.9	-31.1	0.2	-23.8	11.3	-11.3	10.2	-13.0	4.1	320	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	27.0	19.1	24.9	17.4	23.0	16.5	20.5	24.9	19.3	23.2	18.0	21.4	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.6	23.2	17.6	13.5	21.3	17.0	13.0	20.4	61.4	25.2	57.0	23.0	53.3	21.7	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.4	7.4	DB	-31.3	30.0	2.4	1.6	-33.0	31.2	-34.4	32.1	-35.7	33.0	-37.4	34.1	(4)
(5)			WB	-31.1	22.4	2.5	1.6	-32.9	23.6	-34.4	24.5	-35.8	25.4	-37.6	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-12.8	-11.1	-4.8	1.9	8.7	13.7	16.8	15.9	11.7	5.0	-2.5	-9.7	(6)	
	DBStd	11.66	7.16	6.88	6.85	5.13	4.62	3.64	3.36	3.32	4.40	4.73	5.58	6.89	(7)	
	HDD10.0	3282	706	591	460	247	81	8	1	1	32	168	376	612	(8)	
	HDD18.3	5717	964	824	717	493	300	144	71	91	203	413	626	871	(9)	
	CDD10.0	655	0	0	1	3	41	119	212	183	82	13	1	0	(10)	
	CDD18.3	47	0	0	0	0	2	5	23	14	3	0	0	0	(11)	
	CDH23.3	343	0	0	0	1	16	41	169	94	19	2	0	0	(12)	
	CDH26.7	40	0	0	0	0	1	3	22	12	2	0	0	0	(13)	
(14) Wind	WSAvg	3.1	3.6	3.5	3.3	3.4	3.2	2.7	2.5	2.4	2.6	3.1	3.3	3.5	(14)	
(15) Precipitation	PrecAvg	679	29	20	29	50	68	92	84	71	78	84	51	35	(15)	
	PrecMax	908	61	48	83	189	122	212	204	148	173	222	100	75	(16)	
	PrecMin	519	7	4	9	8	20	10	34	22	20	32	14	7	(17)	
	PrecStd	108	15	12	17	38	25	45	43	30	36	47	24	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	6.2	14.0	19.1	25.9	27.1	28.7	27.9	26.2	21.2	12.4	4.6	(19)	
		MCWB	0.5	2.6	9.2	10.4	15.6	19.3	19.8	20.1	17.1	14.6	8.4	2.3	(20)	
	2%	DB	1.0	2.8	9.8	15.9	22.5	24.1	27.2	26.1	22.7	17.5	9.1	2.7	(21)	
		MCWB	-0.6	1.1	5.5	8.2	13.8	16.2	19.4	18.7	17.1	12.0	6.6	1.0	(22)	
	5%	DB	-1.2	1.2	7.1	12.6	19.8	22.6	25.2	23.8	21.1	14.2	7.1	1.2	(23)	
		MCWB	-1.9	-0.4	3.6	6.2	12.4	15.5	18.0	17.5	16.4	10.5	4.5	-0.1	(24)	
	10%	DB	-2.7	-1.2	3.8	10.3	17.3	21.1	23.5	22.4	18.8	12.5	5.1	-0.8	(25)	
		MCWB	-3.7	-2.5	1.0	4.9	10.9	14.9	17.2	16.6	15.0	9.3	2.8	-1.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	3.0	10.7	12.4	17.9	20.6	22.6	22.1	20.1	15.8	10.1	2.8	(27)	
		MCDB	2.7	4.5	13.0	16.6	21.5	25.2	26.8	26.3	23.6	18.5	11.6	4.0	(28)	
	2%	WB	-0.8	1.6	6.4	9.0	15.4	18.7	20.8	20.0	18.2	13.4	7.0	1.3	(29)	
		MCDB	0.3	3.0	9.0	14.4	19.5	22.4	25.2	24.5	20.9	16.4	8.7	2.6	(30)	
	5%	WB	-2.0	-0.5	3.5	6.8	13.6	17.1	19.5	18.8	16.9	11.2	4.7	-0.1	(31)	
		MCDB	-1.4	1.1	6.3	11.7	18.1	20.1	23.5	22.6	19.6	14.1	6.8	1.1	(32)	
	10%	WB	-3.8	-2.5	1.4	5.1	11.6	15.6	18.2	17.6	15.3	9.5	3.0	-1.2	(33)	
		MCDB	-2.7	-1.6	4.0	9.7	16.2	19.4	21.9	20.7	18.4	12.4	4.8	-0.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	9.5	9.6	10.6	10.5	10.6	10.4	9.7	7.8	6.9	7.7	(35)	
		MCDBR	8.3	8.7	10.6	14.0	14.9	12.9	12.0	12.1	12.3	11.2	8.7	7.4	(36)	
	5% WB	MCWBR	7.3	7.0	7.3	8.5	7.9	7.3	6.7	6.9	7.6	7.5	6.6	6.6	(37)	
		MCWBR	7.7	7.8	9.9	12.8	12.2	10.5	10.8	10.8	10.1	10.0	8.2	7.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.245	0.254	0.286	0.325	0.342	0.363	0.372	0.358	0.342	0.301	0.276	0.251	(40)	
		taud	2.434	2.410	2.428	2.425	2.458	2.416	2.433	2.490	2.508	2.594	2.558	2.466	(41)	
	Ebn at Noon		862	928	943	929	921	899	885	885	869	864	819	809	(42)	
		Edh at Noon	63	81	95	106	107	112	109	99	88	69	57	55	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.21	2.23	3.51	4.70	5.39	5.78	6.02	5.11	3.74	2.23	1.21	0.92	(44)	
	RadStd		0.15	0.25	0.32	0.48	0.36	0.42	0.28	0.36	0.25	0.27	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		-1.02	-2.27	N/A	N/A	N/A	N/A	+465	+413	N/A	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page